



Royal
Glass



Ward

Glass Balustrade System

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Mini Post Glass Balustrade
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS (tick boxes that apply)

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ please specify	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ please specify			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS (tick boxes that apply)

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	0.5-1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)

Note: If barrier extends more than 1.2m above fixing point, test reports must be provided

Supplier:	Royal Glass
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PART 4: BARRIER DETAILS

System name:	Mini Post Glass Balustrade
Drawings (ref page number)	
Engineers name:	PNP Engineering
Hardware (type):	316 Stainless Steel

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)

- ☐ Continuous interlinking rail (show how fixed to building and each pane of glass)
- ☐ Continuous interlinking top cap (show how fixed to building and each pane of glass)
- ☐ Clamps, brackets or point fixings (show how fixed to building and each pane of glass)
- ☐ Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)
- ☐ Glass with stiff interlayer
- ☐ Two or three glass edges supported by continuous clamps
- ☐ Structural post to fix interlinking rail to deck
- ☐ Other (specify)

Engineers name:

PNP Engineering

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz



*Completely-frameless look achieved using laminated glass

Ward is a modern solution that combines style and practicality. This system is designed to make your space look great while keeping things simple and safe. No need to worry about water and debris gathering around – the glass are lifted, letting them escape easily. The glass balustrade system can be top-mounted or side-mounted.

2 Choose your glass configuration



Toughened Glass SAFE

Toughened glass have greater strength and safety features than float glass, allowing the glass to break with dull edges.

Interlinking rail and gap clamps may be required*



Laminated Glass

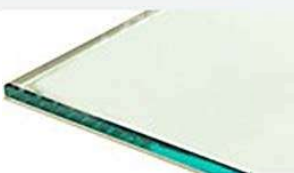
Laminated glass have EVA films between 2 or more layers of toughened glass creating a single panel, guaranteeing that if the glass shatters, it remains bonded together.



SentryGlas™

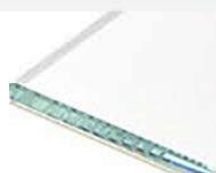
SentryGlas™ have interlayers in the glass that is significantly stronger and stiffer than traditional laminates, ensuring minimal deflection in case of glass breakage.

2 Choose your glass type



Standard

Has a slight green tint from standard glass attributes



Low-Iron

Provides a clearer and more transparent view



Frosted

Gives superior privacy and security



Tinted

Minimises sun exposure and provides privacy

Your balustrades may consist of the following hardware fixings:



A Interlinking Rail



B Glass Clamp



C Rail Joiner



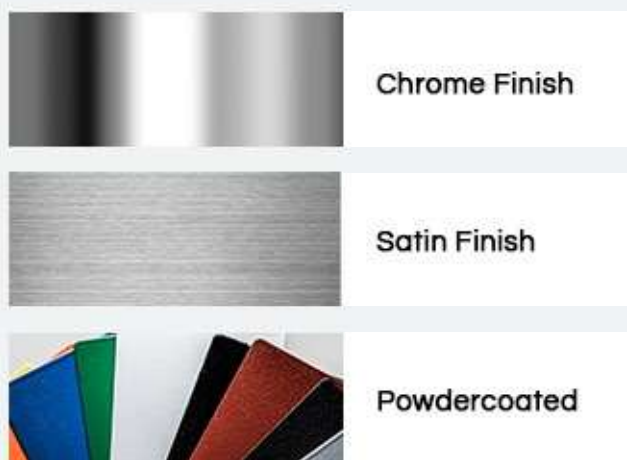
D Wall Clamp



Would you like guidance on the most suitable option for your property? Our team is ready to assist you.

Give us a call at **0800 769 254** or send us an email at info@royalglass.co.nz

3 Choose your fixing colour



P & P CONSULTING ENGINEERS LTD

Civil and Structural Engineering

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Mr J. Dela Cruz
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Ref: 20/059/dd.ds
05/07/2024

ROYAL GLASS LTD

CANTILEVER GLASS BALUSTRADES SUPPORTED BY MINI POST

DESIGN CALCULATIONS SUMMARY

INDEX	PAGE
1. <u>Producer Statement - Design</u>	1
2. <u>Occupancy Types</u>	2
The balustrade has been tested to comply with domestic barriers serving one or more dwelling including balcony edges (occupancy types A & C3 of AS/NZS 1170).	
3. <u>Test Results</u>	3-5
Refer test results on pages 3 – 5. 12 mm toughened glass is assessed to be sufficient for max. height 1.08 m for balustrade, 1.26m for pool fence. Glass to have continuous top rail 50 x 1.5 CHS or 25x21 RHS	
4. <u>Wind Load Assessment</u>	2
The balustrade & pool fence are sufficient for up to “Very High” wind zone.	
5. <u>Fixings</u>	6
Refer to summary drawing ENG 01 & ENG 02.	
6. <u>Interlinking Rail</u>	7-10
Use 50.8 OD x 1.5 mm thick CHS rail or 25 x 21 SHS rail.	
7. <u>Summary</u>	11
Refer summary sketch drawing ENG 01 & ENG 04.	

Notes:

1. Any parts of the structure which are not covered by the specific design included with these calculations must comply either with the New Zealand Building Code or specific design as detailed by others. Any exceptions to this should be referred back to this Design Office.
2. The above calculations include structural work for which a Building Consent must be obtained prior to building. It is the Owner's responsibility to obtain all necessary consents.
3. It is assumed that the strength and stiffness of the substrate is sufficient to adequately resist the balustrade loads – this must be confirmed for each installation situation.
4. This design assumes that all the specified members are suitably protected from excess moisture in accordance with Section E1, E2 and E3 of the Building Code. All timber, steelwork, bolts and fasteners to be corrosion protected in accordance with the requirements of NZS 3604:2011 Chapter 4, Durability.

SUPPLIER NAME: Royal Glass

NAME:

ADDRESS:

DATE:

CONSENT AUTHORITY: Auckland Council

engineering
new zealand



New Zealand
Institute of Architects
Incorporated



THIS PS1 DOES NOT CONSTITUTE A SITE SPECIFIC REVIEW BY THE ENGINEERING COMPANY BELOW AND IS ONLY APPLICABLE FOR STANDARD FIXINGS AND DESIGN PARAMETERS AS DETAILED IN THE RELEVANT MANUAL Building Code Clause(s) B1, F2, F4

PRODUCER STATEMENT – PS1 – DESIGN

(Guidance on use of Producer Statements (formerly page 2) is available at www.engineeringnz.org)

ISSUED BY: P & P CONSULTING ENGINEERS LTD

(Design Firm)

TO: ROYAL GLASS LTD

(Owner/Developer)

TO BE SUPPLIED TO: VARIOUS COUNCIL

(Building Consent Authority)

IN RESPECT OF: GLASS BALUSTRADE & POOL FENCE SUPPORTED BY MINI POSTS, CAPPING RAILS

(Description of Building Work)

AT: VARIOUS

(Address)

Town/City: LOT DP SO

(Address)

We have been engaged by the owner/developer referred to above to provide:

STRUCTURAL DESIGN

(Extent of Engagement)

services in respect of the requirements of Clause(s) B1, F2, F4 of the Building Code for:

☐ All or ☒ Part only (as specified in the attachment to this statement), of the proposed building work.

The design carried out by us has been prepared in accordance with:

☒ Compliance Documents issued by the Ministry of Business, Innovation & Employment. B1/VM1, F2/AS1, F4/AS1 or (verification method/acceptable solution)

☐ Alternative solution as per the attached schedule.

The proposed building work covered by this producer statement is described on the drawings titled:

FRAMELESS GLASS BALUSTRADE & POOL FENCE and numbered 20/059 ENG 01 - ENG 05, together with the specification, and other documents set out in the schedule attached to this statement.

On behalf of the Design Firm, and subject to:

- (i) Site verification of the following design assumptions REFER NOTES AT END OF DESIGN SUMMARY
(ii) All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that a) the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the attached schedule, will comply with the relevant provisions of the Building Code and that b), the persons who have undertaken the design have the necessary competency to do so. I also recommend the following level of construction monitoring/observation:

☐ CM1 ☐ CM2 ☐ CM3 ☐ CM4 ☐ CM5 (Engineering Categories) or ☒ as per agreement with owner/developer (Architectural)

I, PARMIL PRAKASH am: ☒ CPEng 251801 # ☐ Reg Arch #
(Name of Design Professional)

I am a member of: ☒ Engineering New Zealand ☐ NZIA and hold the following qualifications: BE (Civil), CPEng

The Design Firm issuing this statement holds a current policy of Professional Indemnity Insurance no less than \$200,000*.

The Design Firm is a member of ACENZ: ☐

SIGNED BY PARMIL PRAKASH (Signature)
(Name of Design Professional)

ON BEHALF OF P & P CONSULTING ENGINEERS LTD Date 31/10/2024
(Design Firm)

Note: This statement shall only be relied upon by the Building Consent Authority named above. Liability under this statement accrues to the Design Firm only. The total maximum amount of damages payable arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in contract, tort or otherwise (including negligence), is limited to the sum of \$200,000*.

This form is to accompany Form 2 of the Building (Forms) Regulations 2004 for the application of a Building Consent.
THIS FORM AND ITS CONDITIONS ARE COPYRIGHT TO ACENZ, ENGINEERING NEW ZEALAND AND NZIA

P & P CONSULTING ENGINEERS LTD

Civil and Structural Engineering

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Ref: 20/059

TO WHOM IT MAY CONCERN

Dear Sir/Madam,

B2 COMPLIANCE

Producer Statement for Design-PS1 for Clause B2 of the Building Code – Structural Durability.

We are not able to provide this because there is no effective verification method for B2 contained within the Building Code.

However we can confirm that for the structural elements shown on our documentation:

Timber

Timber treatment has been selected in accordance with Table 1A of B2/AS1.

Concrete

Concrete cover have been selected in accordance with NZS3101:2006, Part 1 Section 3.

Mild Steel

Steel protection has been specified in accordance with the “Guide to the protection of structural steel against atmospheric corrosion by the use of protective coatings” AS/NZS3212. We Note that this is on a time to first maintenance basis.

We trust this provides the information that you are seeking.

Yours faithfully,



Mr P Prakash
Director
P & P Consulting Engineers Limited

CANTILEVER GLASS BALUSTRADES TESTING
SUPPORTED BY 5/S ROUND MINI POSTS

2. TEST LOADS (round mini post)

The glass and support posts were tested to comply with the following domestic load cases taken from AS/NZS 1170.1:2002 and the DBH Practice Advisory # 10:

Occupancy Type	Refer Table 3.3 of AS/NZS 1170:	Maximum Design Loads		
A, C3 (Residential Only)	Domestic Barriers for One or More Dwellings including Balcony Edges (NOT subject to Over Crowding)	0.75 kN/m (75 kg/m)	0.6 kN (60 kg) Anywhere	1 kPa (100 kg/m ²) Infill

3. TEST RESULTS (round mini post)

As noted above, the test loads were determined from Table 3 of AS/NZS 1170:2002

3 samples were tested which passed. Refer to test results attached on pages 3 -5

Based on above testing, 12 mm toughened safety glass is sufficient for the occupancy types noted above.

Testing 1

Point :	
Stress(kg)	Distance(mm)
0	0
20	14
40	28
60	45
80	62
100	80
110	91
120	101

Face:	
Stress(kg)	Distance(mm)
0	0
20	2
40	7
60	11
80	16
100	21
120	27
140	33
160	38
180	44
200	49
220	54
240	60
260	65
280	71
300	77
320	82
340	87
360	91

Line:	
Stress(kg)	Distance(mm)
0	0
20	11
40	21
60	32
80	43
100	54
120	65
140	75
160	86
180	97
200	114
220	126
240	139
260	156
280	fracture

Testing 2

Point :	
Stress(kg)	Distance(mm)
0	0
20	15
40	29
60	44
80	59
100	74
110	85
120	95

Face:	
Stress(kg)	Distance(mm)
0	0
20	4
40	8
60	13
80	18
100	23
120	29
140	35
160	40
180	46
200	51
220	56
240	62
260	67
280	73
300	78
320	83
340	88
360	92

Line:	
Stress(kg)	Distance(mm)
0	0
20	8
40	23
60	36
80	48
100	66
120	88
140	98
160	113
180	124
200	136
220	144
240	155
260	165
280	fracture

5

Testing 3

Point:	
Stress(kg)	Distance(mm)
0	0
20	15
40	29
60	44
80	59
100	74
110	85
120	95

Face:	
Stress(kg)	Distance(mm)
0	0
20	4
40	8
60	13
80	18
100	23
120	29
140	35
160	40
180	46
200	51
220	56
240	62
260	67
280	73
300	78
320	83
340	88
360	92

Line:	
Stress(kg)	Distance(mm)
0	0
20	10
40	22
60	36
80	48
100	61
120	73
140	86
160	98
180	111
200	124
220	136
240	150
260	166
280	184
300	200
320	212
340	fracture

5 FIXINGS (round minl post)

Lever Arm = 60 mm for 4 bolts

Line Max BM per splgot = $0.75 \times 1.5 \times 1.0 \times 0.75 = 0.84 \text{ kNm}$

Wind $1.16 \times 1.3 \times 0.5 \times 1.22^2 \times 0.75 \text{ m} = 0.84 \text{ kN}$

Loads per Bolt = $0.84 / 0.06 \times 2 = 7.0 \text{ kN}$

Anchor Type	Capacity (kNt)	Comments
<u>1. To Concrete</u>		
M10 Chemset	21.4 kN	$f_c = 25 \text{ MPa}$, 90 mm edge dist. Epcon C6, $f_c = 25 \text{ MPa}$, embedment = 90 mm
<u>2. To Steel</u>		
M10 Grade 4.6	25 kN	
<u>3. To Timber</u>		
M10 Bolts	25 kN	
Capacity of 50 x 60 washers	22 kN (dry) 13 kN (wet)	Close OK
M10 Coach screws (wet timber)	7.86kN (wet)	Req Length = 150 mm
M10 Coach screws (dry timber)	7.49kN (dry)	Req Length = 100 mm

Testing (circular tube)

Point:	
Stress(kg)	Distance(m)
0	0
20	8
40	22
60	34
80	49
100	61
120	72
150	85

Point:	
Stress(kg)	Distance(m)
0	0
20	9
40	23
60	35
80	48
100	62
120	71
150	86

Point:	
Stress(kg)	Distance(m)
0	0
20	10
40	21
60	33
80	46
100	63
120	74
150	88

8

Testing (square tube)

Point:	
Stress(kg)	Distance(m)
0	0
20	11
40	38
60	57
80	77
100	95
110	108
120	116

Point:	
Stress(kg)	Distance(m)
0	0
20	12
40	35
60	55
80	75
100	96
120	107
150	119

Point:	
Stress(kg)	Distance(m)
0	0
20	11
40	36
60	58
80	76
100	98
120	110
150	125

6. Capping rail

Three glass panel, each panel 2.0m maximum
Say if middle glass panel failure, then
horizontal load transfer to side glass (critical)
or side glass panel failure, the horizontal load
transfer to wall framing structure & middle glass (not critical)

Check top rail member

$$UDL^{\#} = 0.75 \text{ kN/m} \times 2\text{m} = 1.5 \text{ kN} = 150 \text{ kg} \quad (\text{SLS}) \quad \text{critical}$$

$$P^{\#} = 0.6 \text{ kN} = 60 \text{ kg} \quad (\text{SLS})$$

From testing, the top rail can take 160 kg
the rail is not failure (OK)
the connection to glass panel is fixed (OK)

$$\text{Hence top rail capacity} = 160 \text{ kg} / 2\text{m} = 80 \text{ kg/m} > 75 \text{ kg/m} \quad (\text{OK})$$

Therefore use 50.8x15 RHS or 25x21 RHS as capping rail.

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Civil and Structural Engineering
6A Montel Avenue, Henderson ph: 036-1053

Project: Radisthale (SL2)
Designed: H. Yin

Job No.

Sheet of

Date

check end connection to Raillings

2x20 channels. capacity = $0.3 \times 2014 \times 0.3 \times 2 = 2.1 \text{ kN}$

Design load = $0.25 \text{ kN/m} \times 6 \times 2.0 \text{ m} / 2 = 0.75 \text{ kN}$, Hence (OK)

or 140 LMB Drush screw with 30 mm penetration

MS mesh screw capacity = $0.3 \times 0.3 \times (0.5 \times 30 \text{ mm} \times 2.0 \text{ m} \times 3 \text{ mm})$
= $2.02 \text{ kN} > \text{Design load (OK)}$

check bottom fixing

$R^* = 0.25 \text{ kN/m} \times 2 \text{ m} / 2 = 0.25 \text{ kN}$

$M^* = 0.25 \text{ kN} \times 1.0 = 0.25 \text{ kN.m}$

$T^* = 0.25 \text{ kN/m} \times 60 \text{ mm} / 2 = 6.25 \text{ kN}$

Anchor types

① M10 bolts changed into galvanized with 90 mm embedment

$\phi T_{n \text{ channel}} = 2.4 \text{ kN} > T^* \text{ (OK)}$

② M10 A4-6 bolts to steel member

$\phi T_n = 25 \text{ kN} > T^* \text{ (OK)}$

③ M10 bolts with 50x50x3 washer to timber

$\phi T_n = 7.4 \text{ kN} > T^* \text{ (OK)}$

④ M10 coach screw with 150 mm penetration into wet timber or 100 mm into dry timber.

$\phi T_{n \text{ wet}} = 7.36 \text{ kN} > T^* \text{ (OK)}$ $\phi T_{n \text{ dry}} = 7.49 \text{ kN} > T^* \text{ (OK)}$

11. SUMMARY

In summary, the panel & fixings tested conform to the following:

LOADS:

Live Load: For Domestic Occupancy types A and C3 (residential only) of AS/NZS 1170:2002, Table 3.3

Wind Load: HIGH as per NZS 3604 or 44 m/s Ultimate Limit State Wind Speed
HIGH wind zone for Pool Fence

GLASS:

Thickness: 12 mm Toughened Grade A

Maximum height is 1.08 m above finished floor level , 1.22 m for pool fence

Capping Rail: 50.8 OD x 1.5 CHS or 25x21 RHS

FIXINGS:

Refer to attached summary drawing ENG01 to ENG 04.

Note that it is assumed that the strength and stiffness of the substrate is sufficient to adequately resist the balustrade loads .

FRAMELESS GLASS BALUSTRADES SUPPORTED BY MINI POSTS

END BUILDING
(OR 50.8 OD x 1.5 CHS POST)

2000 MAX. WITH 3 MINI POSTS

CAPPING RAIL DETAILS REFER TO SHEET ENG 03 & ENG 04

CAPPING RAIL FOR BALUSTRADE ONLY

CONTINUOUS CAPPING RAIL FOR BALUSTRADE

12 mm TOUGHENED GLASS

1080 MAX. (1260 for POOL FENCE)

THIS GLASS BALUSTRADE SYSTEM IS SUITABLE FOR THE FOLLOWING:

LIVE LOAD:

OCCUPANCY TYPE	REFER TABLE 3.3 OF AS/NZS 1170	DESIGN LOAD
A C3 (Residential Only)	DOMESTIC BARRIERS SERVING ONE OR MORE DWELLINGS INCLUDING BALCONY EDGES. (NOT SUBJECT TO CROWD LOADINGS)	0.6 kN CONC LOAD ANYWHERE, OR 0.75 kN/m LINE LOAD AT 0.9 m ABOVE FLOOR LEVEL

WIND LOAD:

"VERY HIGH" WIND ZONE FOR BALUSTRADE, "VERY HIGH" WIND ZONES FOR POOL FENCE.

FIXINGS:

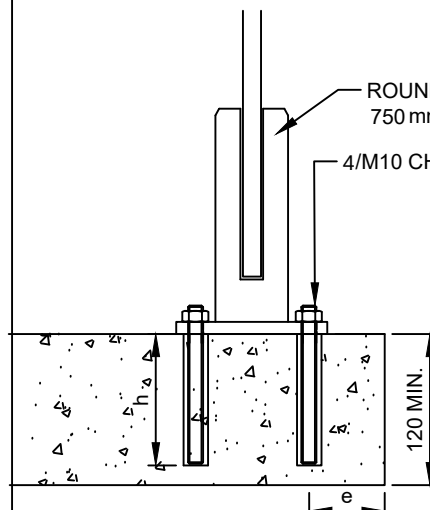
A. ALL ANCHORS MUST BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS REQUIREMENTS.
B. BOLTING ARRANGEMENT MUST COMPLY WITH RELEVANT MATERIAL CODES.
C. STRENGTH & STIFFNESS OF THE SUBSTRATE STRUCTURE MUST BE SUFFICIENT FOR THE BALUSTRADE LOADS.

1. M10 CHEMSET ANCHORS (MAXIMA SPIN CAPSULES OR EPCON C6 SERIES) THE ANCHORS MUST COMPLY FULLY WITH THE MANUFACTURER'S REQUIREMENTS.
2. M10 G4.6 BOLTS TO STEEL MEMBERS. THE BOLTING ARRANGEMENT MUST COMPLY WITH NZS 3404:1997.
3. M10 G4.6 BOLTS TO TIMBER WITH 50 mm x 50 mm SQUARE WASHERS. THE BOLT ARRANGEMENT MUST COMPLY WITH NZS 3603:1993.
4. M10 COACH SCREWS TO WET TIMBER OR DRY TIMBER EMBEDMENT 'H' REFER TO TABLE.

DURABILITY

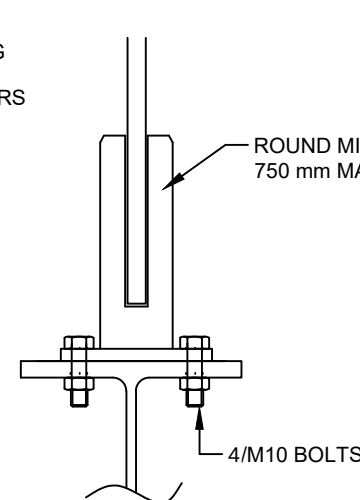
THE FIXINGS MUST COMPLY WITH SECTION 4 "DURABILITY" OF NZS 3604:2011

ELEVATION

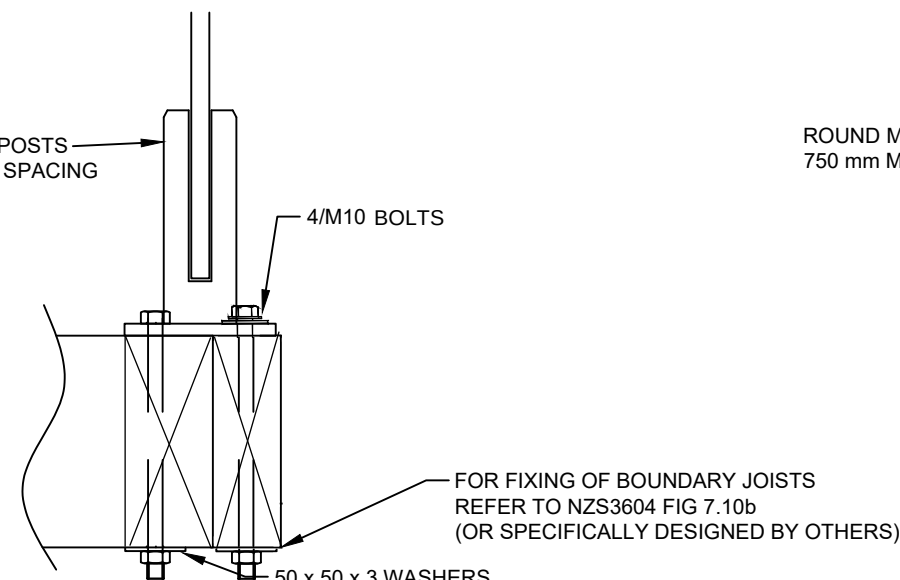


FOR CHEMSET: h = 90 mm, e = 40 mm

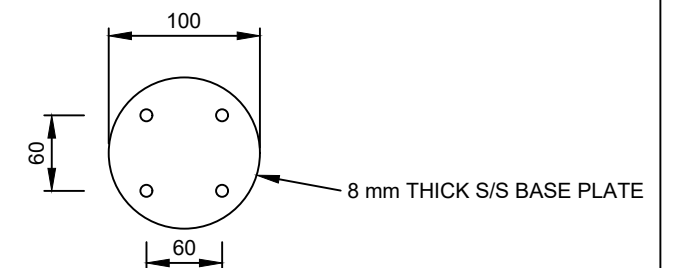
CONNECTION TO CONCRETE



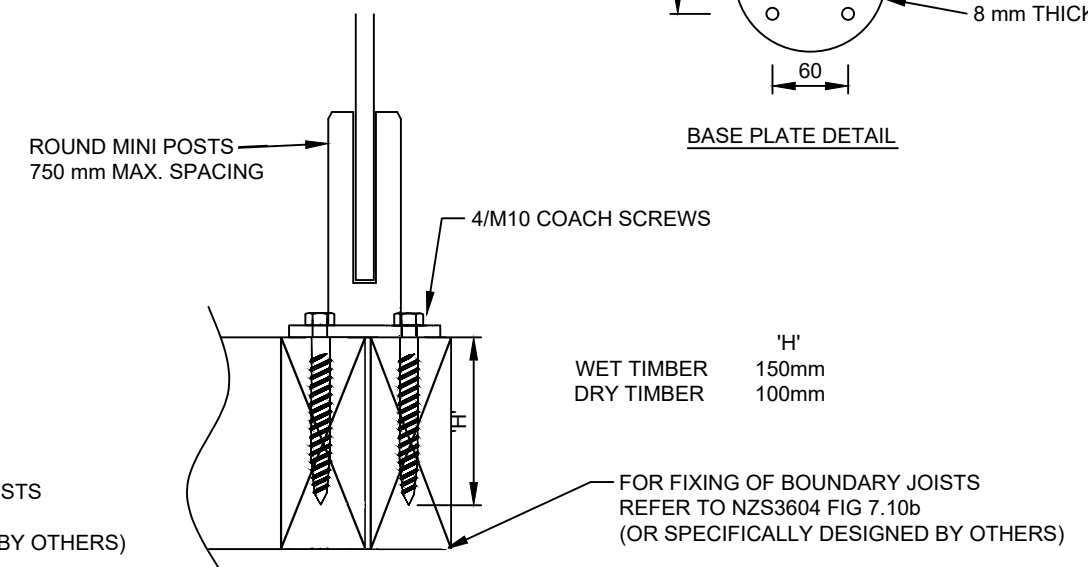
CONNECTION TO STEEL



CONNECTION TO TIMBER



BASE PLATE DETAIL



CONNECTION TO TIMBER

P & P CONSULTING ENGINEERS LTD

Civil and Structural Engineering
6A Montel Avenue, Henderson ph: 836-1853

ROYAL GLASS LTD

FRAMELESS GLASS BALUSTRADES

GLASS BALUSTRADE SUPPORTED BY MINI POSTS

Scales 1:10 (A3)

Drawn

Date 30/06/2020

Project

Sheet ENG 01

FRAMELESS GLASS BALUSTRADES SUPPORTED BY MINI POSTS

END BUILDING
(OR 50.8 OD x 1.5 CHS POST)

1500 MAX. WITH 2 MINI POSTS

CAPPING RAIL DETAILS REFER TO SHEET ENG 03 & ENG 04

CAPPING RAIL FOR BALUSTRADE ONLY

CONTINUOUS CAPPING RAIL FOR BALUSTRADE

12 mm TOUGHENED GLASS

1080 MAX. (1260 for POOL FENCE)

THIS GLASS BALUSTRADE SYSTEM IS SUITABLE FOR THE FOLLOWING:

LIVE LOAD:

OCCUPANCY TYPE	REFER TABLE 3.3 OF AS/NZS 1170	DESIGN LOAD
A C3 (Residential Only)	DOMESTIC BARRIERS SERVING ONE OR MORE DWELLINGS INCLUDING BALCONY EDGES. (NOT SUBJECT TO CROWD LOADINGS)	0.6 kN CONC LOAD ANYWHERE, OR 0.75 kN/m LINE LOAD AT 0.9 m ABOVE FLOOR LEVEL

WIND LOAD:

"VERY HIGH" WIND ZONE FOR BALUSTRADE, "VERY HIGH" WIND ZONES FOR POOL FENCE.

FIXINGS:

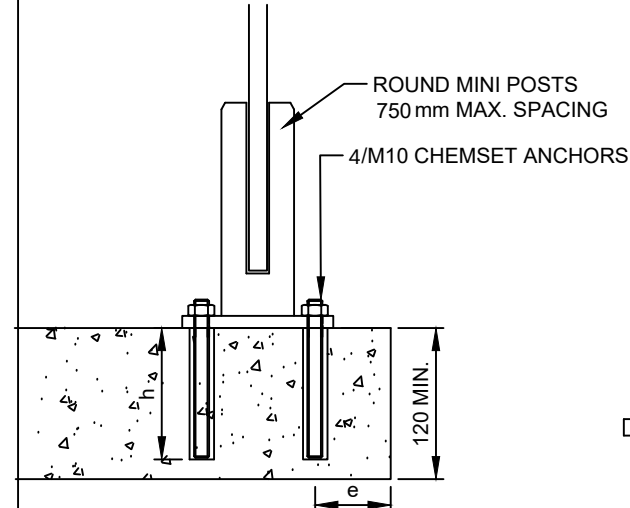
A. ALL ANCHORS MUST BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS REQUIREMENTS.
B. BOLTING ARRANGEMENT MUST COMPLY WITH RELEVANT MATERIAL CODES.
C. STRENGTH & STIFFNESS OF THE SUBSTRATE STRUCTURE MUST BE SUFFICIENT FOR THE BALUSTRADE LOADS.

1. M10 CHEMSET ANCHORS (MAXIMA SPIN CAPSULES OR EPCON C6 SERIES) THE ANCHORS MUST COMPLY FULLY WITH THE MANUFACTURER'S REQUIREMENTS.
2. M10 G4.6 BOLTS TO STEEL MEMBERS. THE BOLTING ARRANGEMENT MUST COMPLY WITH NZS 3404:1997.
3. M10 G4.6 BOLTS TO TIMBER WITH 50 mm x 50 mm SQUARE WASHERS. THE BOLT ARRANGEMENT MUST COMPLY WITH NZS 3603:1993.
4. M10 COACH SCREWS TO WET TIMBER OR DRY TIMBER EMBEDMENT 'H' REFER TO TABLE.

DURABILITY

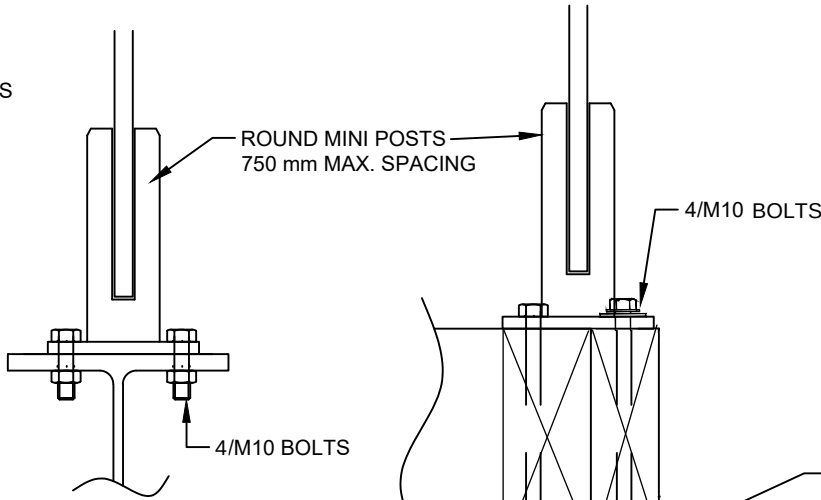
THE FIXINGS MUST COMPLY WITH SECTION 4 "DURABILITY" OF NZS 3604:2011

ELEVATION

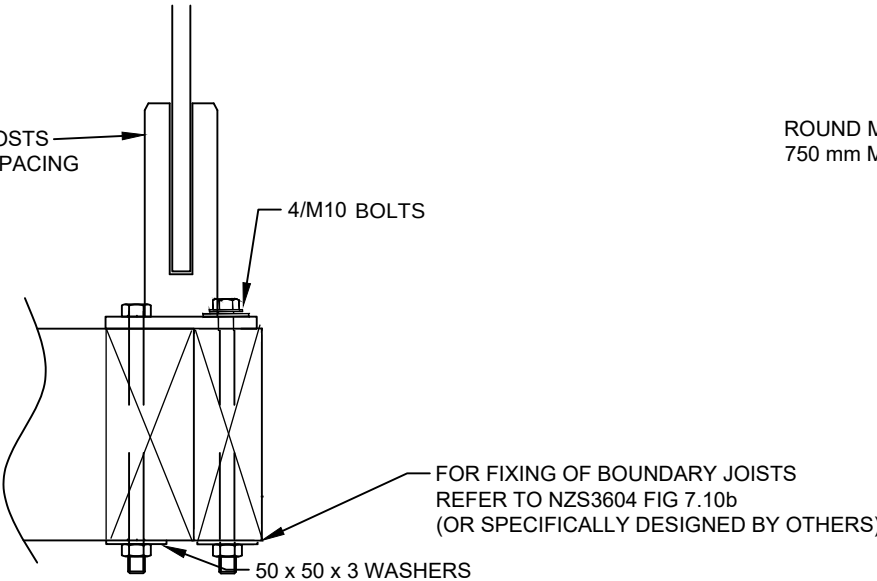


FOR CHEMSET: h = 90 mm, e = 40 mm

CONNECTION TO CONCRETE

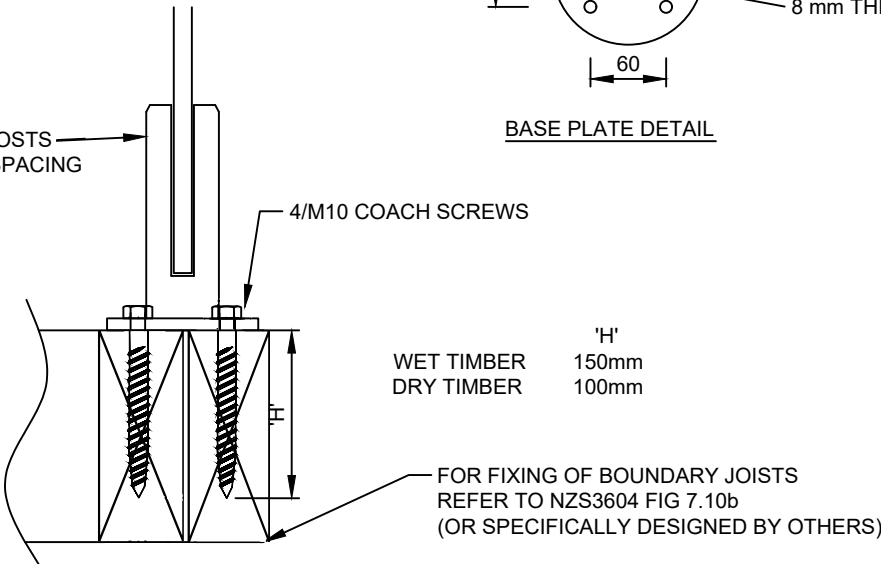


CONNECTION TO STEEL

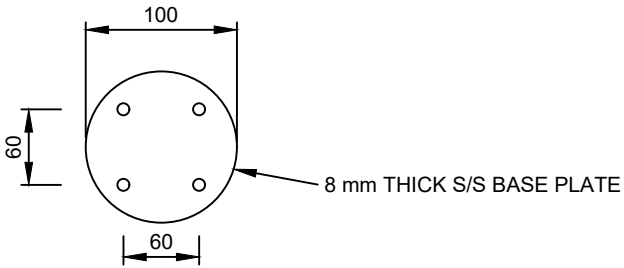


CONNECTION TO TIMBER

ROUND MINI POSTS
750 mm MAX. SPACING



CONNECTION TO TIMBER



BASE PLATE DETAIL

P & P CONSULTING ENGINEERS LTD

Civil and Structural Engineering
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ROYAL GLASS LTD

FRAMELESS GLASS BALUSTRADES

GLASS BALUSTRADE SUPPORTED BY MINI POSTS

Scales 1:10 (A3)

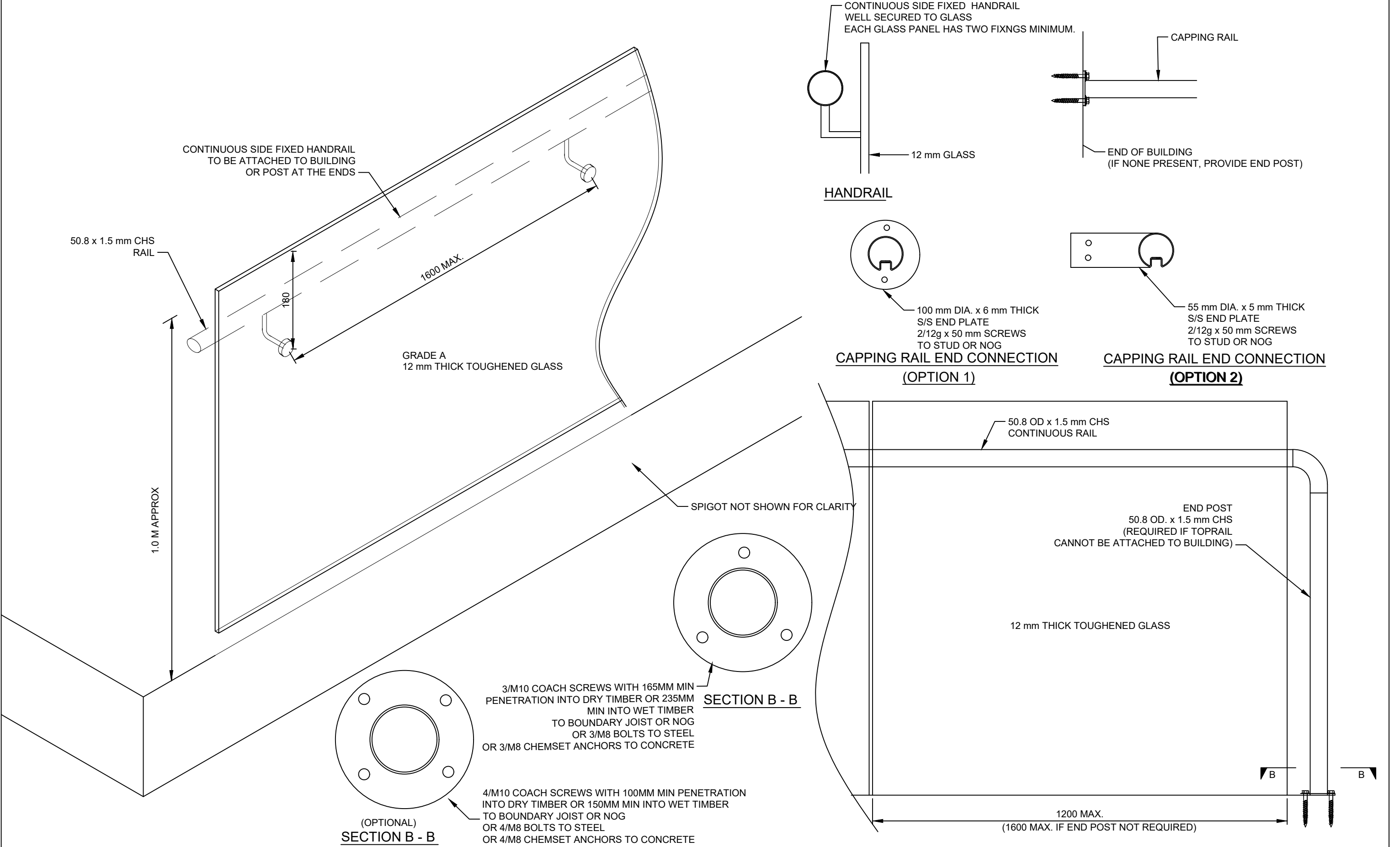
Drawn

Date 30/06/2020

Project

Sheet ENG 02

GLASS BALUSTRADES CAPPING RAIL (SIDE FIXED)



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FRAMELESS GLASS BALUSTRADES

GLASS BALUSTRADE SUPPORTED BY MINI POSTS

Scales 1:10 (A3)

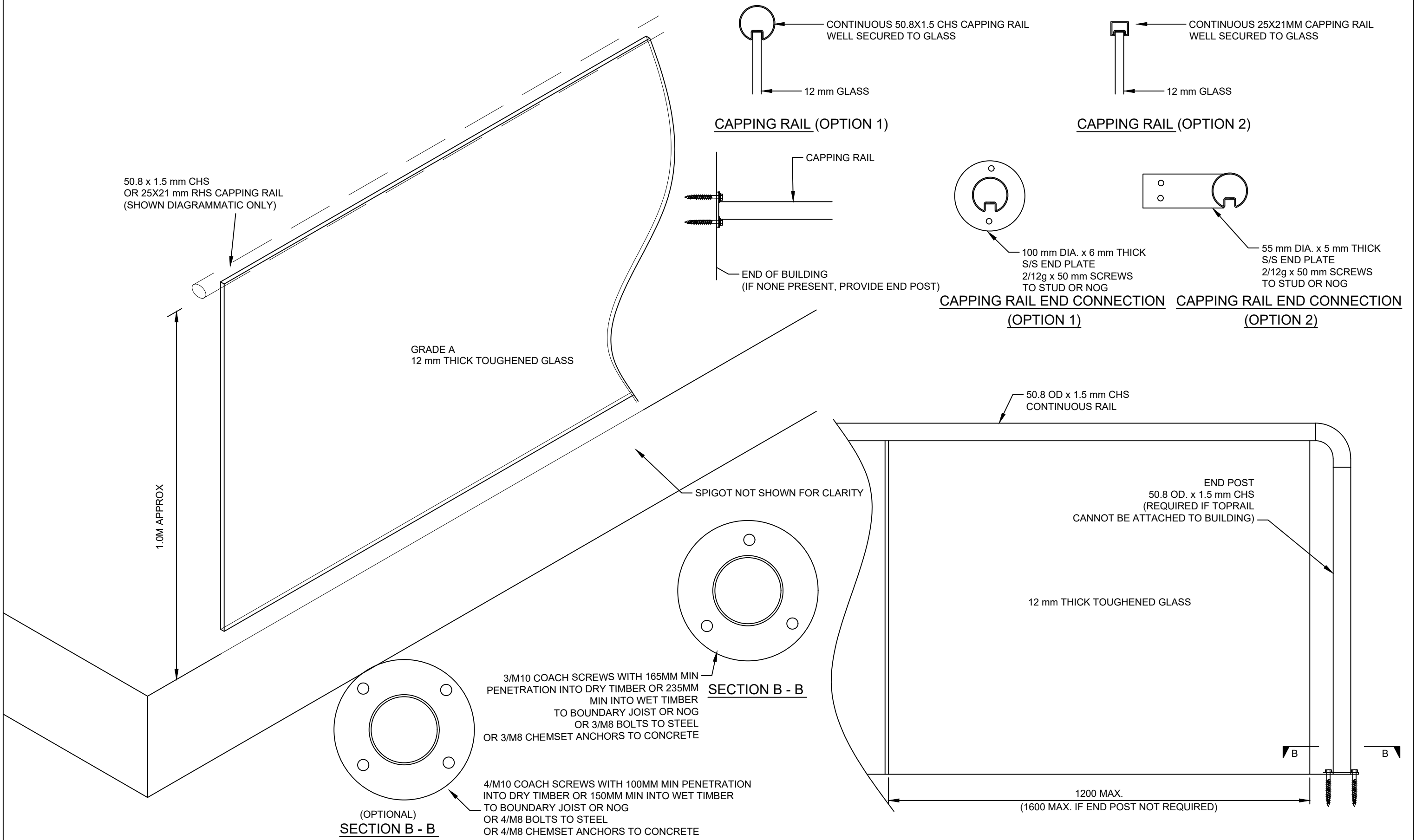
Drawn HY

Date 30/06/2020

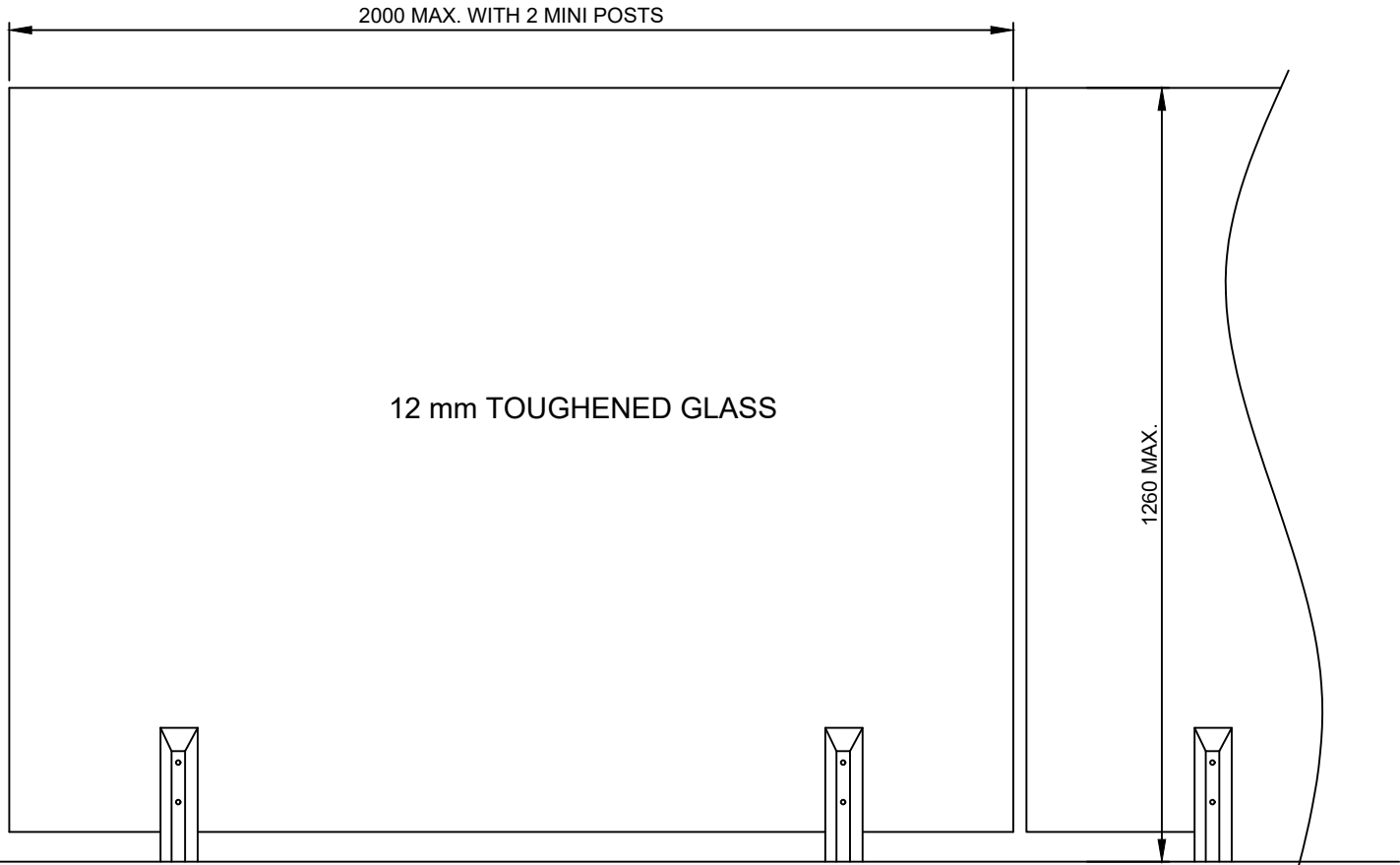
Project

Sheet ENG 03

GLASS BALUSTRADES CAPPING RAIL (TOP FIXED)



FRAMELESS GLASS POOL FENCE SUPPORTED BY MINI POSTS

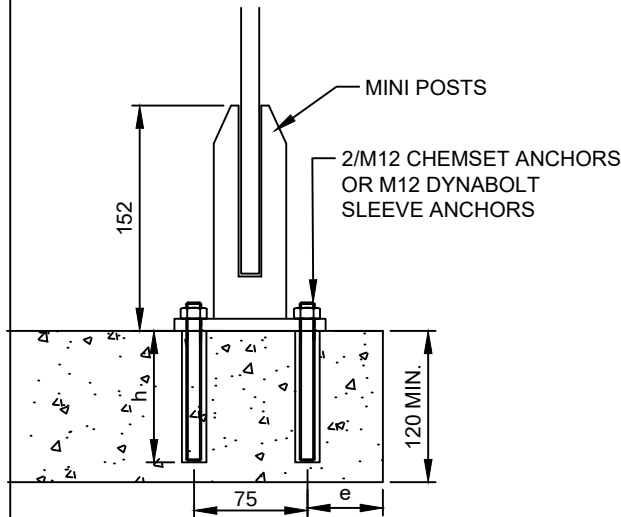


WIND LOAD: "VERY HIGH" WIND ZONE .

- FIXINGS:
1. ALL ANCHORS MUST BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS REQUIREMENTS.
 2. BOLTING ARRANGEMENT MUST COMPLY WITH RELEVANT MATERIAL CODES.

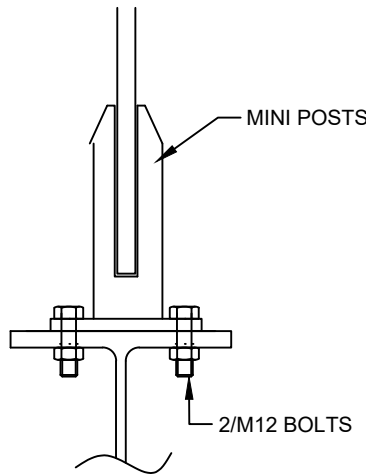
DURABILITY
THE FIXINGS TO BE STAINLESS STEEL.

ELEVATION

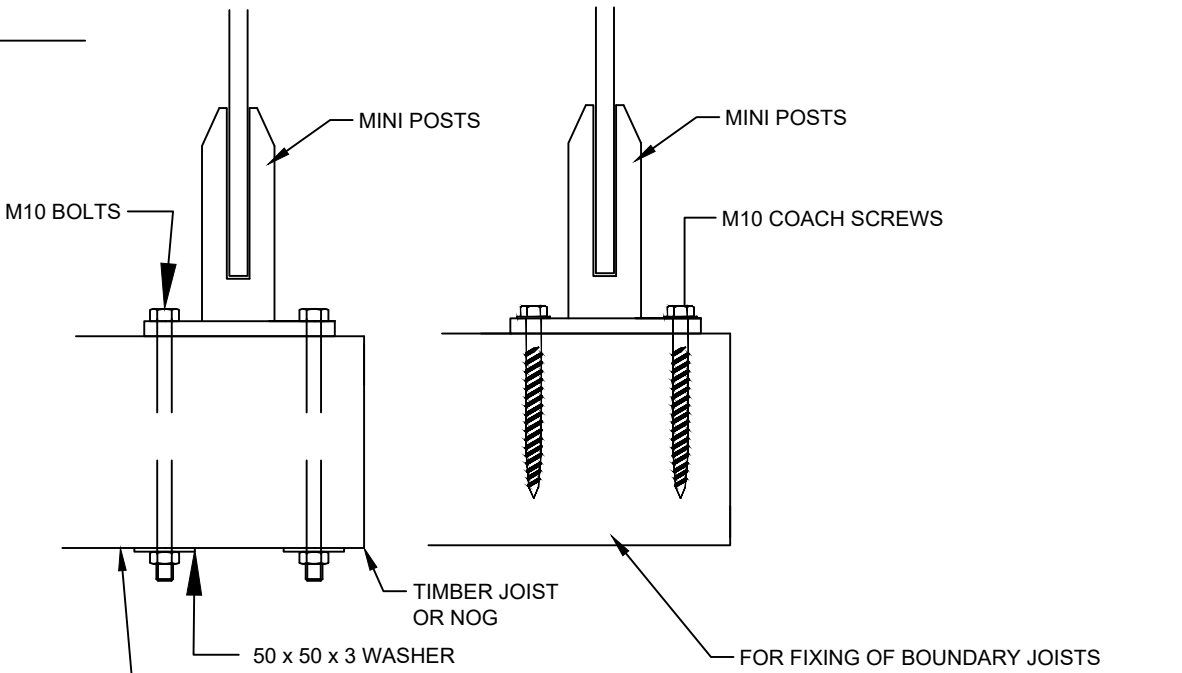


FOR CHEMSET: h = 90 mm, e = 40 mm
FOR DYNABOLT: h = 70 mm, e = 90 mm

CONNECTION TO CONCRETE



CONNECTION TO STEEL



CONNECTION TO TIMBER

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ROYAL GLASS LTD FRAMELESS GLASS POOL FENCE

GLASS POOL FENCE SUPPORTED BY MINI POSTS

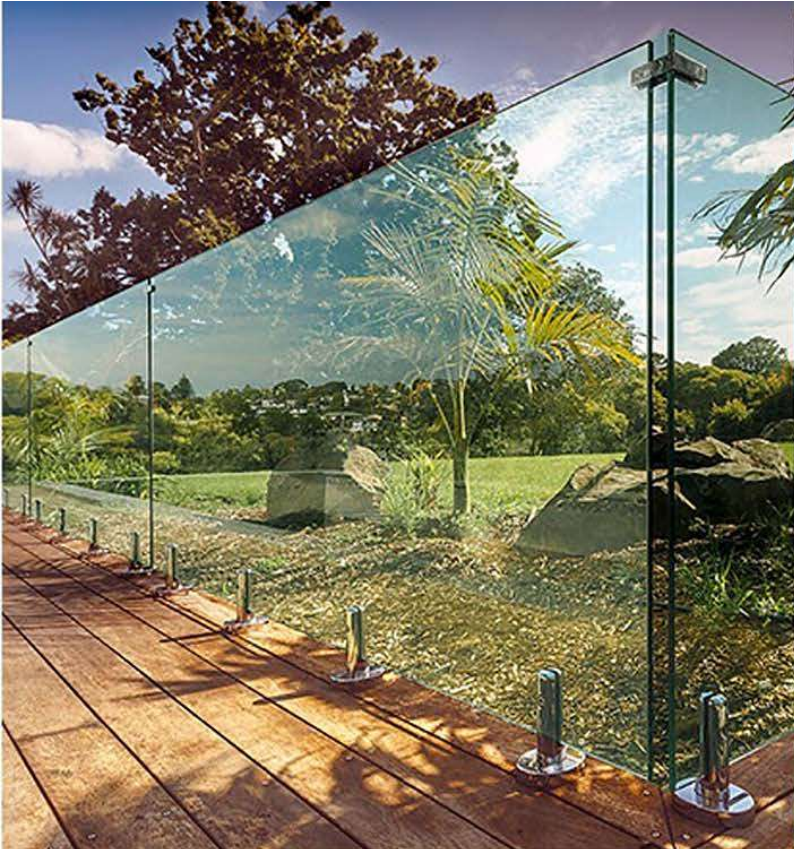
Scales 1:10

Drawn PP

Date

Project

Sheet ENG 05



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